Indirect reciprocity with inertial reputation updating

Introduction, Storyline 2: accurate agreement from private revision

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,8,18]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[12,19,20]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[18,21,22]. The dilemma can be eased in two ways. Observers can be brought to agree about each other's standing, so that justified punishment is recognized as such, or reputations can be made robust enough that a single disputed judgement does not overturn them. In both cases, genuine defectors must still lose their standing.

Most remedies take the first route and align what observers know. Public institutions broadcast a single verdict[23], gossip spreads opinions between peers[24,25], and other mechanisms synchronize private opinions[17] or evaluate actions from the donor's perspective[20]. Many of them rely on empathy, communication or institutional trust, and they are strained where these are compromised, as in polarized societies[21,26]. Agreement can also emerge from private information alone, when observers aggregate several observations of the same individual, keep quantitative scores or assess recipients as well as donors[21,27,28]. In many settings, evidence about a particular individual arrives piecemeal, one encounter at a time, and observers carry their judgements from one encounter to the next. Agreement then depends on how each observer revises its judgement as evidence accumulates. Such agreement is useful only if it is also accurate, because observers who regard almost everyone as bad agree with one another without distinguishing cooperators from defectors. Which properties of individual revision yield agreement that is also accurate, and in particular whether losing and regaining standing should follow the same rule, remains unclear.

Here, we develop a model of indirect reciprocity in which each observer revises its private judgements with inertia, and in which this inertia can differ between losing and regaining standing (Fig. 1). Observers share no information, and each update draws on a single observation. An established judgement changes only after a run of contrary assessments. Negative assessments move an individual in good standing through a sequence of warning states before the observer's verdict turns bad, and positive assessments move an individual in bad standing through recovery states before it turns good. Each intermediate state

is interpreted as good or bad with a probability that sets how strongly incomplete evidence already affects the observer's behaviour.

We find that private observers reach agreement that is also accurate, although they share no information, provided that inertia is allocated in a norm-specific direction. With the same small inertia in both directions, observers under Shunning agree only because they regard nearly everyone as bad. Compact architectures, selected solely by agreement and accuracy, make cooperation stable and easy to reach under each of four classical norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all, so how evidence is revised matters more than how much of it is retained. Finally, the tolerance that makes private verdicts robust can also shelter opportunists who defect only occasionally, a vulnerability that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Agreement about reputations therefore need not be imposed through shared information, but whether it is accurate depends on how each individual revises its judgements.

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